ANDROID HANDHELD LAUNCHER · FEATURE OVERVIEW

Pavo.

Everything on the device,
under the thumbs.

Version 1.0.42

Quick Controls edition

PACKAGE	PLATFORM	REFERENCE DEVICE	LICENCE
com.ben.pavo	Android 12 or newer	AYN Odin 3 · Android 15	Apache 2.0

November 2026 · github.com/Ben0019/pavo-launcher

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Feature areas

A complete replacement for the stock launcher, written against the Android platform with no AndroidX, no Gradle runtime and no root-only hooks.

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NEW IN 1.0.42 · QUICK CONTROLS

- **Edge grab handle** — a slim accent bar pinned to any screen edge.
- **Swipe inwards** to slide the panel over the running app.
- **Long-press & drag** the handle to park it on the left, right, top or bottom edge.
- **BRI/VOL** vertical sliders that write live while you drag.
- **PERF / FAN / REF** cards use the tuner and fan-control colours.
- **SHOT · REC · MAP** actions, with long-press variants.
- **Region capture** — drag, move, resize, then save or reset.
- **Single-app recording** from the REC long press.
- **Per-app mapper** — MAP binds the touch map to the app behind the panel.

ALSO IN THIS BUILD

Handle width, height, opacity and visibility are adjustable from Overlay & Drawer, and the handle is hidden from screenshots and screen recordings while a capture runs. Region capture saves to Pictures/Pavo Screenshots with a _region suffix. Fan HOLD now anchors to the selected preset's peak clock and runs the blower at the set speed, cooldown ramps are smoothed, and the SMALL, MEDIUM and LARGE fan curves are quieter at the same temperatures. The performance overlay adds a live FAN duty row. All new strings ship in the 25 launcher languages.

Ten bands over everything.

A system-wide graphic equalizer built on Android's DynamicsProcessing effect. Every app shares one curve, and edits take effect immediately — no Apply step.

Mixer

- **31 Hz to 16 kHz** in ten bands, each with a ±12 dB vertical slider.
- **0 dB centre line** shows boost or cut at a glance.
- **Drag anywhere** on a slider, or use the right stick on the focused band.
- **0.1 dB** right-stick steps for fine adjustment.

Presets & lock

- **27 presets** in a compact 9×3 grid, all visible without scrolling.
- **PAVO** stores the current curve as its default; long-press PAVO to save a new one.
- **LOCK** freezes the bands. Presets still apply, save and persist.
- **EQ ON/OFF** sits beside LOCK; the controller focus ring follows the knob, not the column.

CURVE MIGRATION

Existing five-band curves are redistributed onto the ten-band grid once, by log-frequency interpolation. The five-value presets are expanded automatically, so saved tunings keep their shape.

Calibrate the sticks. Keep the drift out.

A built-in calibration and testing surface for the Odin sticks, with live raw and corrected positions, a gate map and per-app profiles.

Calibration

- **Guided** rest, range and gate capture; centre and reset are separate.
- **RAW / OUT** preview with colour-matched legend dots.
- **Deadzone, CURVE and Min output** sliders, applied live.
- **LINEAR, SMOOTH, PRECISE, FAST, RACING, FPS and ACCEL** presets; long-press to reset.
- **EXPERT** unlocks the wider deadzone and curve ranges.

Per-app & controller

- **Per-app profiles** follow the game; AUTO switches, MANUAL keeps the global tuning.
- **BYPASS** stops the remap daemon for that app and exposes the physical controller.
- **Left stick / D-pad** navigate; **A** activates; **B** closes.
- **Right stick** adjusts the focused slider one step at a time, with each step saved.
- **Hold A** performs long-press actions.

CONTROLLER IDENTITY

The virtual pad advertises itself as an Xbox Wireless Controller. When a game rejects synthetic controllers, turn on that app's BYPASS: the daemon stops, the virtual pad disappears and the game sees the physical controller only.

One swipe from the edge. Everything, over the game.

A compact glass card that floats above the running app without pausing it. Brightness, volume, performance, fan and refresh rate sit in one reachable column, with capture actions along the bottom.

Panel

- **BRI / VOL** — tall vertical sliders. Both write while you drag: no release step.
- **PERF** — SMALL / MEDIUM / LARGE / STOCK with the tuner's own profile colours.
- **FAN** — SMALL, MEDIUM, LARGE, HOLD, OFF, SMART, HIGH, CURVE in a 2×4 grid, coloured like Fan Control.
- **REF** — Auto, 60 Hz and 120 Hz.
- Chips wrap tightly, titles are centred, the card is sized to the two-column grid.

Actions

- **SHOT** — full screenshot. **Long press** opens the region selector.
- **REC** — whole-screen recording through the built-in recorder, started invisibly. **Long press** records a single app.
- **MAP** — opens the in-game Touch Mapper overlay only, never the mapper app.
- Mapping is bound to the app behind the panel: enabling or editing from here saves to that app's profile.

REGION CAPTURE

- **Drag** a rectangle over the frozen screen.
- **Move** the box from inside, or drag any of the eight handles to resize.
- **Live size badge** shows the exact pixel crop.
- **SAVE · DELETE · RESET · CANCEL** bar, styled like the recorder HUD.
- The bar **dodges the selection**, hopping to the other edge.
- Saved to Pictures/Pavo Screenshots with a media scan and toast.

LAYOUT

Row 1	BRI/VOL card (104 dp) + PERF card (78 dp)	Panel width	Exactly two columns plus padding — 212 dp
Row 2	FAN card (104 dp) + REF card (78 dp)	Actions	Three equal buttons along the bottom
Card height	150 dp for every preset card	Labels	BRIT · VOL · PERF · FAN · REF, centred and tracked

Park the handle anywhere.

The grab handle is a small accessibility overlay that stays above other apps. It never appears in a screenshot or a recording, and it snaps back to its saved slot after a restart.

Handle

Placement	Right, left, top or bottom edge. Long-press, drag, release to snap.
Open	Swipe inwards — right edge: left, left edge: right, top: down, bottom: up.
Size	Width 2–16 dp, length 40–400 dp.
Opacity	10–100 %, live preview.
Visibility	Show or hide it entirely from the same card.
Saved	Edge and slot are remembered; re-applied on every reboot.

Overlay & drawer card

The old overlay actions card is replaced by **QUICK CONTROLS**: a show/hide toggle and three sliders for the grab handle. The metrics card gives up room so the column still fits without clipping.

Gesture summary

- Tap a control to act; long-press where a variant exists.
- The panel dismisses itself before a capture so it never appears in the image.
- Handle touches are excluded from the system back gesture.

BEHAVIOUR

- The panel is a trusted accessibility overlay, so games that hide non-system overlays still show it.
- Brightness and volume write through the privileged channel and the audio service; both are throttled to about 50 ms while dragging.
- The mapper overlay opened from MAP keeps its own per-app layout — buttons, sticks, combos and zones are stored against the package.

SETTINGS CARD

SHOW HANDLE	Adds or removes the handle overlay immediately.	Opacity	10–100 %, live preview while sliding.
Handle width	2–16 dp, applied to the bar and its touch strip.	Metrics card	Gives up column share so the new card fits.
Handle height	40–400 dp; the bar re-centres on the current edge.	Saved	All four values persist and re-apply on restart.

Two rings. Two zones. Full control.

The Odin 3 exposes one SN3112 LED driver per joystick. Each stick has a single live RGB zone, so the launcher paints two independent colours at once — one per hand — with every mode, colour and gesture available from the RGB app.

What the hardware can do

- One addressable zone per stick; the second logical zone is not wired on the Odin 3.
- Eight driver channels all address the same physical zone — no gradient around a single ring.
- Two sticks, two colours simultaneously, each with its own mode.

Modes

- **STATIC** fixed colour · **BREATH** speed-controlled pulse
- **RAINBOW** hue rotation · **STROBE** on/off flashes
- **MULTI** two hues, 120° apart · **SCREEN** follows the display
- **OFF** disables that stick but keeps its colour

Per-stick editing

- **BOTH** shares one colour and mode; switching adopts the selected stick's settings.
- **EACH** gives each stick its own colour and mode; long-press toggles one stick off without forgetting its mode.
- Editable hex field: #RRGGBB, RRGGBB and #RGB, with type, delete, copy, paste and cut. Invalid codes restore the last valid colour.
- Controller: D-pad or left stick moves focus, A selects, hold A toggles the focused stick.

SCREEN mode

- Left half of the display drives the left stick; right half the right stick.
- 160 ms sampling with a 0.38 blend settles a scene change in roughly 0.25 s.
- Dominant hue voting plus 22 % of the plain average keeps neutral areas grey and vivid accents vivid.
- Pauses while the RGB app is open, so it never samples its own UI.

MODE BEHAVIOUR

STATIC	Fixed colour; each stick keeps its own stored colour.
BREATH	Slow brightness pulse; speed controls the cycle, colour is preserved.
RAINBOW	Continuous hue rotation across the spectrum.
STROBE	On/off flashes at the speed setting.
MULTI	Two hues at once, one per stick, both cycling with a 120° offset.
SCREEN	Follows the display; each half drives its stick.
OFF	Disables that stick without forgetting its last colour or mode.

Accuracy choices

- Dominant-hue voting ignores large neutral areas, so a vivid accent is not averaged into grey.
- 22 % of the plain average is mixed back to keep the actual tint.
- No palette cycling: the result is stable while the screen is stable.

What it cannot do

- Show a spatial gradient around one ring — the ring is a single zone.
- Show more than two simultaneous colours across both sticks.
- Capture DRM-protected or secure surfaces.

THE REST OF PAVO

A complete handheld home

01 Home screen

Paged icon grid, folders, resizable widgets, clock/weather/status header, transparent dock, layout backup and restore as a ZIP.

02 App drawer & navigation

Full drawer, dock pop-up drawer, predictive back, install shortcuts, focus-first D-pad navigation.

03 Appearance & wallpaper

Liquid-glass dock, drawers and sheets; wallpaper-derived accent, icon packs, per-surface transparency, OLED protection.

04 Performance tuning

Per-cluster clocks, governors and GPU frequency table; presets and per-app profiles, background hold, thermal guard.

05 Performance overlay & logs

CPU/GPU temperature, clock and load, RAM, FPS graph, per-core rows, battery temperature and power draw.

06 Stress testing

CPU, GPU and combined runs with fixed clocks or a tuning profile; live graph, averages, maxima, throttle and hold figures.

07 Fan control

Off, Low, Smart, High, Curve and Manual; 15-point curve, CPU/GPU/BOTH source, per-app profiles.

08 Device health & charging

Battery model, health, cycles, resistance, USB details; 80 % limit with automatic charge separation.

09 File manager

Dual panes for internal, SD and USB OTG; copy, cut, paste, rename, delete, ZIP, previews and text editing.

10 FTP host & client

Share storage with read/write switches and QR login; mount a remote server as a pane with live previews.

11 RGB lighting

Two independent zones, seven modes, per-stick editing, editable hex, gesture toggles, screen-follow colour.

12 Controllers, overlays & shortcuts

System-wide remapping, per-app layouts, touch mapping, in-game overlay menu and physical shortcuts.

15 **Quick controls**

Edge-handle panel with live brightness and volume, performance and fan presets, refresh rate, screenshots, region capture and recording.

14 Languages, compatibility & limits

25 locales; Android 12+; 1920×1080 and 1280×960 with portrait fallback; clock and fan writes remain firmware requests.

UNDER THE HOOD

- Plain Java against the Android platform: no AndroidX, no Gradle runtime.
- Built-in apps extend one glass activity and share the card material.
- Per-app profiles for performance, fan and touch mappings.
- PServer is the only privileged channel; nothing else asks for root.
- 25 language catalogues are loaded at runtime, no resource rebuild.
- Everything local — no accounts, no telemetry.

CONTROLLER & INSTALL

The ring never leaves you stranded

Home & drawers

- **A** opens the focused app or activates the focused control.
- **Hold A** opens the item menu: rename, icon, info, remove, folder.
- **B** backs out of a menu, folder or drawer.
- **L1 / R1** previous and next page; also page edges in lists.
- **L2** dock pop-up drawer; **R2** full drawer.

Quick controls

- Swipe the edge handle inwards with a finger to open the panel.
- Long-press the handle to drag it to another edge.
- Long-press **SHOT** for a region, **REC** for single-app capture.

RGB & system shortcuts

- **A** on a joystick selects it; **Hold A** toggles it off and on.
- **BTN_C** floating side drawer; hold for its style menu.
- **BTN_Z** performance overlay; hold for the overlay menu.
- Perf Logs key — optional third shortcut.

Getting started

- **01** Install Pavo.apk or run adb install -r Pavo.apk.
- **02** Open Pavo once and grant or skip each permission in the setup sheet.
- **03** Set Pavo as the home app, then press Home.

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RGB ZONES	LIGHTING MODES	LANGUAGES

COMPATIBILITY & LIMITS

- Android 12+ (API 31), 25 languages, 1920×1080 and 1280×960 with portrait fallback.
- Privileged writes use the vendor PServer channel; no su, Magisk or KernelSU dependency.
- One RGB zone per stick: two simultaneous colours maximum; clock and fan writes can still be reduced by the vendor thermal manager.

SHORTCUTS AT A GLANCE

- **Edge handle** — swipe inwards: panel. Long-press + drag: move it.
- **SHOT** tap: screenshot. Long press: region.
- **REC** tap: whole screen. Long press: single app.
- **MAP** tap: in-game mapper overlay for the app behind the panel.
- **BTN_C** drawer, hold for style. **BTN_Z** overlay, hold for its menu.
- **B** closes the panel, the region selector and the mapper editor.

WHERE FILES GO

- Screenshots: Pictures/Pavo Screenshots
- Region crops: …/Aurora_<time>_region.png
- Recordings: Movies/Pavo Recordings
- Single-app clips keep the app_ prefix.
- All captures are media-scanned, so they appear in Gallery immediately.
- Layout backups save as a setup ZIP.